Alexey Koshevoy

Laboratoire de psychologie cognitive et psycholinguistique, ENS-PSL, Postdoctoral researcher
alexeykochevoy@gmail.com | alexeykosh.github.io

Research Interests

Cultural evolution; computational linguistics; linguistic typology; efficient communication; agent-based modelling; strong inference

Education

Ph.D. in Cognitive Science, Aix-Marseille University (France) 2022–2025
Thesis title: "A cultural-evolutionary account of linguistic laws" (defended on 25 of September 2025)
Supervisors: Isabelle Dautriche, Olivier Morin
Committee members: Simon Kirby, Gemma Boleda, Paula Rubio-Fernández, and Benjamin Spector

Diplôme de l'ENS (DENS) in Sciences[1], Ecole Normale Superieure, PSL (France) 2019–2022
Majored in Cognitive Science

M.Sc. in Cognitive Science, Ecole Normale Superieure, PSL (France) 2019–2021
Majored in Linguistics
Thesis title: "Contact-induced changes in colexification patterns"
Supervisors: Thierry Poibeau and Alexandre François

B.A. in Fundamental and Applied Linguistics, Higher School of Economics (Russia) 2015–2019
Majored in Computational Linguistics with a minor in Mathematics
Thesis title: "Observer's paradox: data from the interviews in the Ustja River Basin Corpus"
Supervisor: Michael Daniel

Academic Positions

Postdoctoral Researcher, LSCP, ENS–PSL (France) 2026–...

Doctoral Researcher, CRPN, Aix-Marseille University (France) 2022–2026

Visiting PhD student, CiMEC, University of Trento (Italy) 2024

Visiting PhD student, Centre for Language Evolution, University of Edinburgh (Scotland) 2023

Theses

A. Koshevoy. "A cultural evolutionary account of linguistic laws". PhD thesis. Aix-Marseille University, 2025.

Submitted Publications and Preprints

A. Koshevoy, Sobchuk, and O. Morin. "Slovotvir: Grassroots reform of the Ukrainian language reveals the role of drift and selection in language change." (in revision).

A. Koshevoy and J. Szymanik. "Convexity bias (possibly) makes languages efficient." (in revision). `https://osf.io/preprints/psyarxiv/wtpke_v1`.

[1] Selective, research-focused master's-level program combining advanced coursework and independent research.

A. Koshevoy, M. Halo, R. H. Maudsley, and O. Morin. "Short words are more likely to refer to multiple meanings across 192 language families" (in revision).

A. Koshevoy. "Morphological causatives in Abaza". *Higher School of Economics Research Paper No. WP BRP* 75 (2018).

Peer-reviewed publications

A. Koshevoy, I. Dautriche, O. Morin, and K. Smith. "Coordination, rather than pragmatics, shapes colexification when the pressure for efficiency is low." *Proceedings of the 46th Annual Meeting of the Cognitive Science Society* (2024).

O. Morin and A. Koshevoy. "A Cultural Evolutionary Model for the Law of Abbreviation". *Topics in cognitive science* (2024).

A. Koshevoy, I. Dautriche, and O. Morin. "Why do some words have more meanings than others? A true neutral model for the meaning-frequency correlation." *Proceedings of the 45th Annual Meeting of the Cognitive Science Society* (2023).

A. Koshevoy, H. Miton, and O. Morin. "Zipf's Law of Abbreviation holds for individual characters across a broad range of writing systems". *Cognition* 238 (2023), p. 105527.

A. Koshevoy, A. Panova, and I. Makarchuk. "Building a Universal Dependencies Treebank for a Polysynthetic Language: the Case of Abaza". *Proceedings of the Sixth Workshop on Universal Dependencies (UDW, GURT/SyntaxFest 2023)*. 2023, 1–6.

M. Daniel, A. Koshevoy, I. Schurov, and N. Dobrushina. "Can recall data be trusted? Evaluating reliability of interview data on traditional multilingualism in highland Daghestan". *Field Methods* 34.4 (2022), 288–302.

Talks

Conference Talks

More than 17 talks & poster presentations in international conferences and workshop in 5 countries, including:

- "Short words are more likely to colexify multiple meanings than long words: empirical and experimental evidence", 58th Annual Meeting of the Societas Linguistica Europaea (Bordeaux, France), 2025
- "Slovotvir: Grassroots reform of the Ukrainian language reveals the role of drift and selection in language change", Explaining culture: and interdisciplinary approach (Paris, France), 2025
- "Zipf's Law of Abbreviation holds for individual characters across a broad range of writing systems", Writing & Cognition Workshop (Ascona, Switzerland), 2024
- "Coordination, rather than pragmatics, shapes colexification when the pressure for efficiency is low", Cognitive Science Society Meeting (Rotterdam, Netherlands), 2024
- "Slovotvir: a natural experiment in word replacement", Computational Humanities Research (Paris, France), 2023
- "Can pragmatic reasoning explain the pressure for efficiency in communication?", Internal and external pressures shaping language workshop (Ljubljana, Slovenia), 2023
- "Why do some words have more meanings than others? A true neutral model for the meaning-frequency correlation", Cognitive Science Society Meeting (Sydney, Australia), 2023
- "Frequency and character complexity in 27 writing systems", EVOLANG (Tokyo, Japan), 2022

Invited Talks

- "What makes efficient communication efficient?", Computational Psycholinguistics Laboratory, MIT, 2024
- "How cultural evolution and artificial languages can shed light on the origins of linguistic ambiguity?", CiMEC seminar, University of Trento, 2024
- "Tutorial: From Elan to Universal Dependencies", LIFT, 2020

Teaching Experience

I have taught several courses, including one as the primary instructor. I hold the French "qualification" in language sciences (qualification aux fonctions de maître de conférences, section 07), which enables me to apply for assistant professor positions in France.

Instructor, Introduction to Linguistics for Data Science — Autumn 2024, Spring 2026
Designed and taught an undergraduate course at CPES "Sciences de données, arts et cultures" (32/46 hours, PSL/Lycée Louis-le-Grand)

Guest lecturer, Collegio Clesio seminar — Spring 2024
Invited lecture "Introduction to data-driven linguistics" (2 hours, University of Trento)

Guest lecturer, Language and Cognition — Autumn 2023
Invited lecture and seminar at the Cognitive Science Master (4 hours, Aix-Marseille University)

Academic Service

- Supervised 3 undergrad students and 1 master student at ENS-PSL and Aix-Marseille University
- Reviewer for Cognitive Science, Open Mind, Cognition, Language and Cognition.
- Reviewer (2024, 2025), and meta-reviewer (2026) for the Cognitive Science Society conference.
- Co-organizer of the "Efficiency principles in non-linguistic information transmission" workshop (MPI-EVA, Leipzig, Germany)

Grants and Awards

Max Planck Society Ph.D. scholarship, Max Planck Institute for Geoanthropology	2023
Contrat Doctoral Specifique Normalien, ENS-PSL	2022-2025
International Selection Scholarship, ENS Paris	2019-2022
Increased State Academic Scholarship, Higher School of Economics	2019

Skills

Programming: Python (*Scikit-Learn, Numpy, PyTorch, transformers, BayesFlow, Flask*), R (*ggplot2, lme4, igraph, brms*), Julia, SQL, shell, git, JavaScript, HTML, CSS
Editing: LaTeX, markdown, pandoc
Software: Praat, ELAN, RStudio, PyCharm, Jupyter, VS Code, ChatGPT, GitHub Copilot, MS Office
Languages: Russian (native), English (fluent), French (fluent), German (basic knowledge)